

第6章 沉淀分析法

6.1 溶解度与溶度积

6.2 影响沉淀溶解度的因素

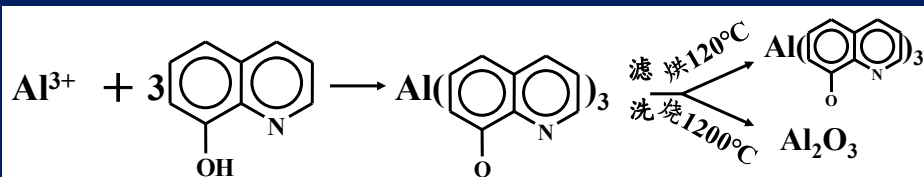
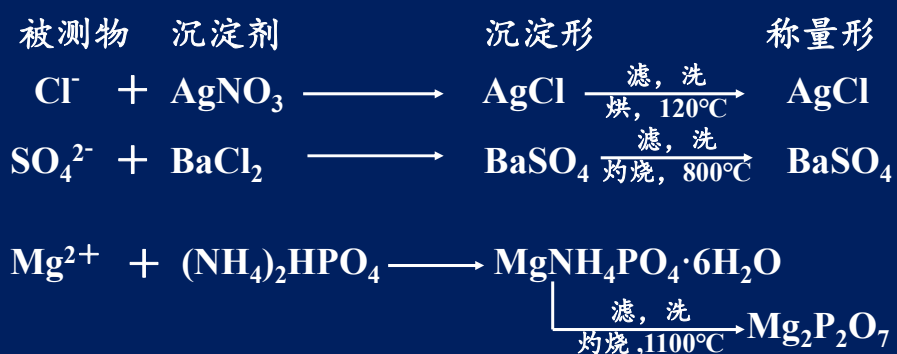
6.3 定量沉淀和分别沉淀

6.4 重量分析

6.5 重量分析计算

6.6 沉淀滴定

重量法示例



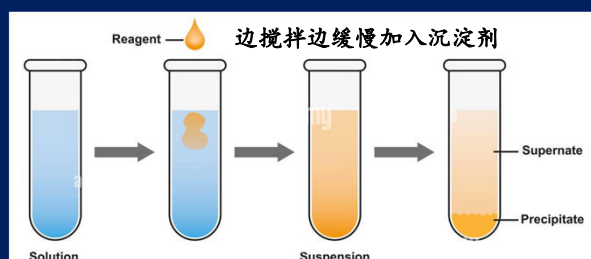
共沉淀小结

- 吸附共沉淀（服从吸附规则）
是胶体沉淀不纯的主要原因
减免方法——洗涤
- 包藏共沉淀（服从吸附规则）
是晶形沉淀不纯的主要原因
减免方法——陈化或重结晶
- 混晶共沉淀
减免方法——预先将杂质分离除去
- 后沉淀
主沉淀形成后，“诱导”杂质随后沉淀下
减免方法——缩短沉淀与母液共置的时间

均匀沉淀法

一般，稀溶液沉淀；边搅拌边缓慢加入沉淀剂

非均匀沉淀

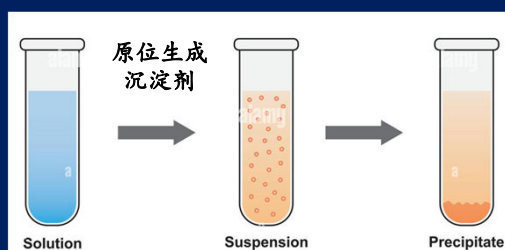


均匀沉淀法

均匀沉淀法：沉淀密度大；沉淀体积小，容易过滤；
沉淀的比表面积小，共沉淀少，结晶纯净

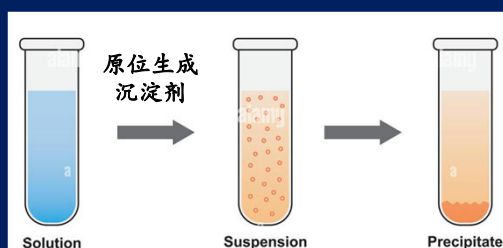
利用酸效应；缓慢产生沉淀剂，课本p146

均匀沉淀

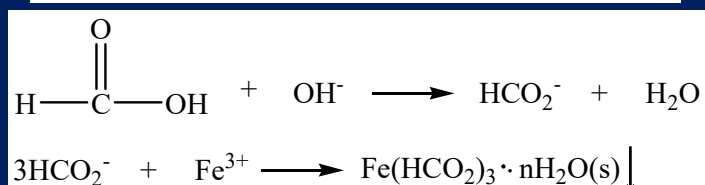
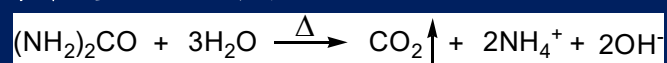


均匀沉淀法

均匀沉淀



例：尿素在沸水中缓慢分解出氢氧根，逐渐增加的氢氧根增大了甲酸铁沉淀的粒径：

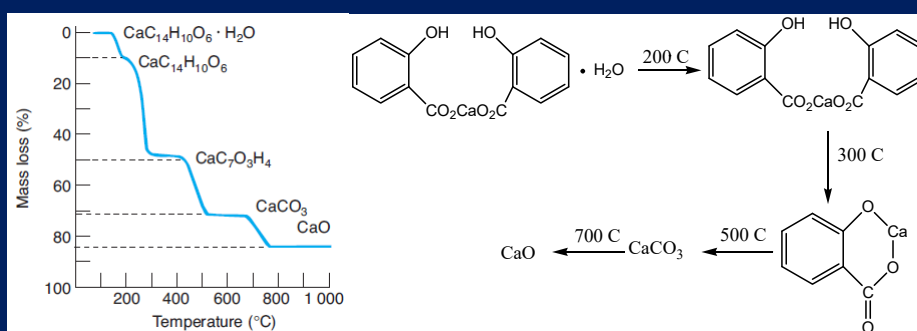


均匀 沉淀 Common reagents used for homogeneous precipitation

Precipitant	Reagent	Reaction	Some elements precipitated
OH^-	Urea	$(\text{H}_2\text{N})_2\text{CO} + 3\text{H}_2\text{O} \rightarrow \text{CO}_2 + 2\text{NH}_4^+ + 2\text{OH}^-$	Al, Ga, Th, Bi, Fe, Sn
OH^-	Potassium cyanate	$\text{HOCN} + 2\text{H}_2\text{O} \rightarrow \text{NH}_4^+ + \text{CO}_2 + \text{OH}^-$	Cr, Fe
S^{2-}	Thioacetamide ^a	$\begin{array}{c} \text{S} \\ \\ \text{CH}_3\text{CNH}_2 \end{array} + \text{H}_2\text{O} \rightarrow \begin{array}{c} \text{O} \\ \\ \text{CH}_3\text{CNH}_2 \end{array} + \text{H}_2\text{S}$	Sb, Mo, Cu, Cd
SO_4^{2-}	Sulfamic acid	$\text{H}_3\text{NSO}_3^- + \text{H}_2\text{O} \rightarrow \text{NH}_4^+ + \text{SO}_4^{2-} + \text{H}^+$	Ba, Ca, Sr, Pb
$\text{C}_2\text{O}_4^{2-}$	Dimethyl oxalate	$\text{CH}_3\text{OCCOCH}_3 + 2\text{H}_2\text{O} \rightarrow 2\text{CH}_3\text{OH} + \text{C}_2\text{O}_4^{2-} + 2\text{H}^+$	Ca, Mg, Zn
PO_4^{3-}	Trimethyl phosphate	$(\text{CH}_3\text{O})_3\text{P}=\text{O} + 3\text{H}_2\text{O} \rightarrow 3\text{CH}_3\text{OH} + \text{PO}_4^{3-} + 3\text{H}^+$	Zr, Hf
CrO_4^{2-}	Chromic ion plus bromate	$2\text{Cr}^{3+} + \text{BrO}_3^- + 5\text{H}_2\text{O} \rightarrow 2\text{CrO}_4^{2-} + \text{Br}^- + 10\text{H}^+$	Pb
8-Hydroxyquinoline	8-Acetoxyquinoline	$\begin{array}{c} \text{O} \\ \\ \text{CH}_3\text{CO} \end{array} \text{C}_8\text{H}_6\text{N} + \text{H}_2\text{O} \rightarrow \begin{array}{c} \text{OH} \\ \\ \text{C}_8\text{H}_6\text{N} \end{array} + \text{CH}_3\text{CO}_2\text{H}$	Al, U, Mg, Zn

^a Hydrogen sulfide is volatile and toxic; it should be handled only in a well-vented hood. Thioacetamide is a carcinogen that should be handled with gloves. If thioacetamide contacts your skin, wash yourself thoroughly immediately. Leftover reagent is destroyed by heating at 50°C with 5 mol of NaOCl per mole of thioacetamide. [H. Elo, J. Chem. Ed. 1987, 64, A144.]

重量分析和热重分析法

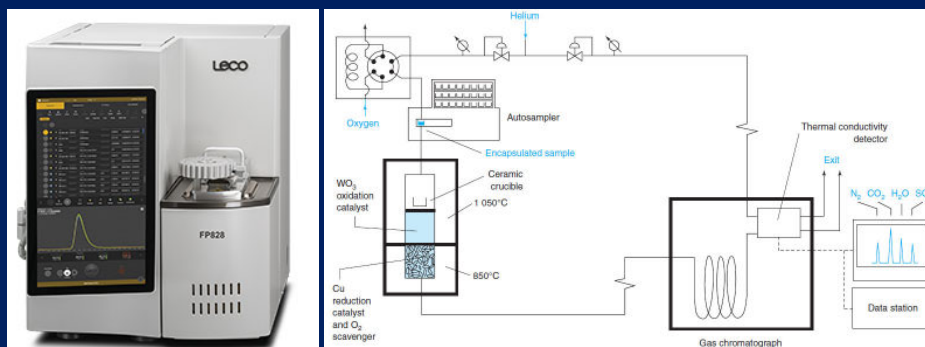


Thermogravimetric curve for calcium salicylate

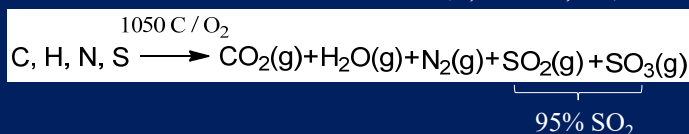
即，最终组成取决于温度和加热时间

有机物元素分析 (elemental analysis by combustion)

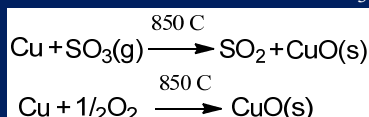
C、H、N、S 元素分析仪，燃烧后通过气相色谱分析 N_2 、 CO_2 、 H_2O 、 SO_2



- 1 准确称量约2 mg样品，密封于锡或银胶囊中
- 2 以预先除去O₂, CO₂, H₂O 的He气充满分析仪
- 3 测量开始，将一定量过量的O₂送入He气流中
- 4 胶囊落于预先加热的陶瓷坩埚中，融化，样品快速氧化



- 5 产物通过热的WO₃氧化催化剂使C完全燃烧生成CO₂；下一区域，还原催化剂金属Cu在850°C将SO₃还原为SO₂并去除过量O₂



- 6 GC分离热导检测。关键是dynamic flash combustion产生短脉冲气体产物流，而非数分钟内缓慢渗出产物。

TABLE 26-4 Accuracy and precision of combustion analysis of pure compounds^a

Substance	C	H	N	S
C ₇ H ₉ NO ₂ S theoretical wt%	49.10	5.30	8.18	18.73
Toluene-4-sulfonamide	49.1 ± 0.63	5.3 ± 0.31	8.2 ± 0.38	18.7 ± 0.89
C ₄ H ₇ NO ₂ S theoretical wt%	36.07	5.30	10.52	24.08
4-Thiazolidinecarboxylic acid	36.0 ± 0.33	5.3 ± 0.16	10.5 ± 0.16	24.0 ± 0.53
Mean uncertainty (wt%) for 7 different compounds	±0.47	±0.24	±0.31	±0.76

a. Results for two of seven pure compounds that were analyzed by 33–45 laboratories each year over six years. Each lab analyzed each compound at least five times during at least two days. For each substance, first row gives theoretical wt% and second row gives measured wt%. Uncertainties are 95% confidence intervals computed for all results after rejecting outliers at the 1% significance level.

SOURCE: R. Companyó, R. Rubio, A. Sahuquillo, R. Boqué, A. Maroto, and J. Riu, "Uncertainty Estimation in Organic Elemental Analysis Using Information from Proficiency Tests," *Anal. Bioanal. Chem.* 2008, 392, 1497.

6.5 重量分析计算

例1：在含 BaCl₂ 的溶液 150.00ml 中加入稍微过量的 AgNO₃，得到 0.7160g AgCl。问(1) BaCl₂ 溶液 150.00ml 中以 BaCl₂ · 2H₂O 形式存在多少克(2) BaCl₂ 溶液的摩尔浓度(3)该 BaCl₂ 溶液 100.00ml 中加入稍微过量的 Na₂SO₄，应得到多少克 BaSO₄ 沉淀？



$$1/2 \text{ AgCl mol} = \text{BaCl}_2 \text{ mol} = \text{BaCl}_2 \cdot 2\text{H}_2\text{O mol}$$

$$\text{AgCl mol} = 0.7160 / 143.3 = 4.997 \times 10^{-3} \text{ mol}$$

$$\text{BaCl}_2 \cdot 2\text{H}_2\text{O g} = 1/2 \times 4.997 \times 10^{-3} \times 244.3 = 0.6104 \text{ g}$$

(2)

$$\text{BaCl}_2 \text{ mol/L} = 1/2 \times 4.997 \times 10^{-3} \times 1000/150 \\ = 1.666 \times 10^{-2} \text{M}$$

(3)

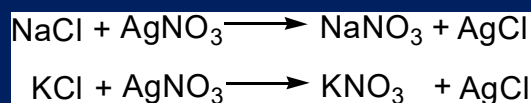


$$\text{BaSO}_4 \text{ mol} = \text{BaCl}_2 \text{ mol}$$

$$\text{BaSO}_4 \text{ g} = 1.666 \times 10^{-2} \times 100 \times 10^{-3} \times 233.4 \\ = 0.3888 \text{ g}$$

例2：将含 NaCl和KCl的样品0.5311g 适当处理后得到1.1519g AgCl。计算样品中含NaCl和KCl的百分率。

设：NaCl g = x
KCl g = y



$$x + y = 0.5311$$

$$\text{AgCl mol} = \text{NaCl mol} + \text{KCl mol}$$

$$\frac{x}{58.44} + \frac{y}{74.55} = \frac{1.1519}{143.32}$$

$$\text{NaCl}(\%) = \frac{x}{0.5311} \times 100 = \frac{0.2470}{0.5311} \times 100 = 46.51\%(W/W)$$

$$\text{KCl}(\%) = \frac{y}{0.5311} \times 100 = \frac{0.2841}{0.5311} \times 100 = 53.49\%(W/W)$$

例3：含Cl⁻的溶液适当处理后得到AgCl。沉淀在过滤、洗涤、干燥、称量过程中由于光分解，使1.50%(w/w)的Ag游离。计算分析结果中产生多少负误差。



假设得到1.0000g沉淀

则：AgCl的量为0.9850g，游离Ag的量为0.0150g

$$\text{AgCl mol} = \text{Ag mol}$$

0.0150g Ag相当的AgCl的质量

$$(0.0150/107.87) \times 143.32 = 0.0199\text{g}$$

无游离Ag时的沉淀质量

$$(0.9850 + 0.0199)\text{g}$$

分析精度

$$\frac{1.0000}{0.9850 + 0.0199} \times 100 = 99.51\%$$

负误差

$$100 - 99.51 = 0.49\%$$

6.6 沉淀滴定

利用沉淀反应建立的滴定方法

银量法 argentometric titrations:



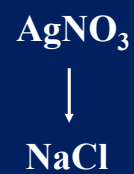
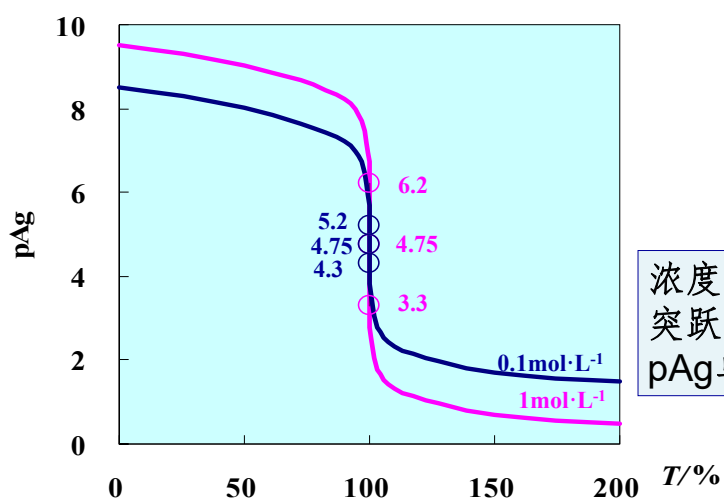
可测 Cl^- 、 Br^- 、 I^- 、 SCN^- 和 Ag^+

滴定曲线

$\text{p}K_{\text{sp}}(\text{AgCl})=9.5$ ($I=0.1$)

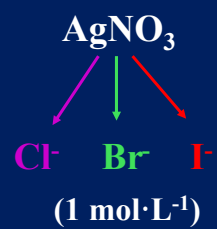
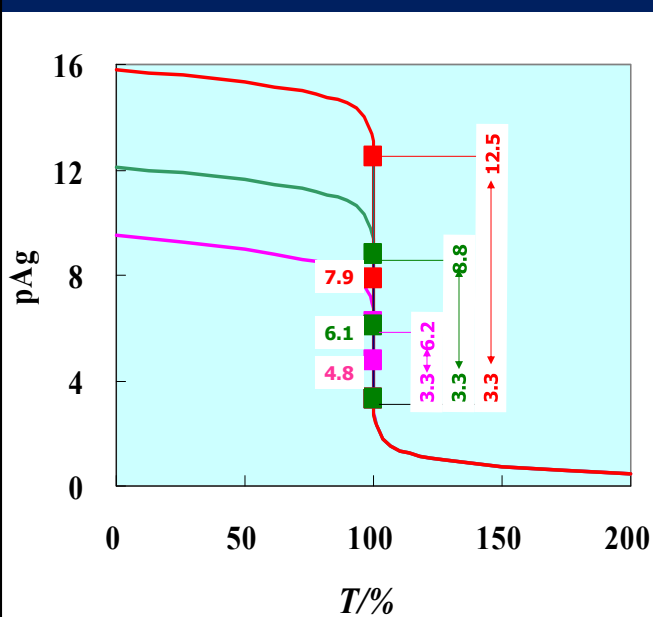
			$\text{AgNO}_3 (0.10 \text{ mol} \cdot \text{L}^{-1})$	
			$\downarrow$	
			$\text{NaCl} (0.10 \text{ mol} \cdot \text{L}^{-1})$	
突跃 {	$T/\%$	pCl	pAg	sp前: 按剩余 Cl^- 计
	0.0	1.0		
	90.0	2.3	7.2	
	99.0	3.3	6.2	
	99.9	4.3	5.2	$\Rightarrow \text{sp}: \sqrt{K_{\text{sp}}}$
	100.0	4.75	4.75	
	100.1	5.2	4.3	sp后: 按过量 Ag^+ 计
	101.0	6.2	3.3	
	110.0	7.2	2.3	
	200.0	8.0	1.5	

滴定曲线



浓度增大10倍,
突跃增加2个
 pAg 单位

滴定曲线



K_{sp} 减小 10^n ,
突跃增加 n
个 pAg 单位

沉淀滴定（银量法）的指示剂

摩尔(Mohr)滴定

K_2CrO_4 为指示剂。 $AgNO_3$ 滴定 Cl^- 或 Br^- ,
砖红色 Ag_2CrO_4 沉淀

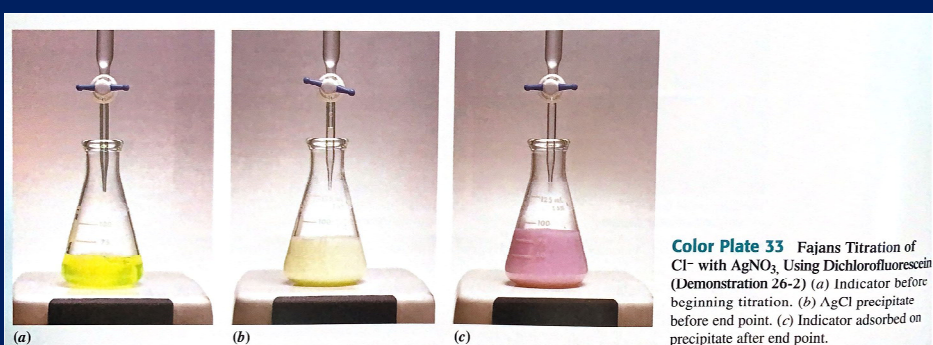
佛尔哈德(Volhard)滴定

Fe^{3+} 为指示剂。含卤素离子的溶液中，加入一定
量过量 $AgNO_3$ ，用 SCN^- 返滴定 Ag^+
红色 $FeSCN^{2+}$

法扬司(Fajans)滴定

指示剂为染料阴离子如荧光素或曙红。化学计量点时，
沉淀表面有稍过量的 Ag^+ ，吸附指示剂阴离子而显色。

Fajans法滴定 Cl^- (荧光黄为指示剂)



滴定前
黄色

滴定中
乳白色

终点后
粉色

以 Ag^+ 滴定 Cl^- 为例

指示剂: 荧光黄(fluorescein)

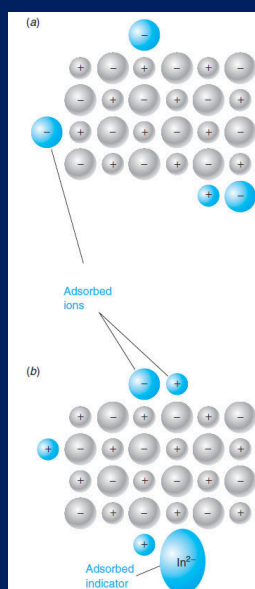
sp前:

$\text{AgCl}:\text{Cl}^-$, 不吸附 Fl^- , 指示剂本身的颜色(黄绿色)

sp后:

$\text{AgCl}:\text{Ag}^+:\text{Fl}^-$, 优先吸附 Ag^+ , Fl^- 作为抗衡离子被吸附, 吸附后结构变形而表现为粉红色

Fajans法 Ag^+ 滴定 Cl^- 沉淀表面终点前后



(a) 终点前, Cl^- 过剩, 沉淀表面负电荷多

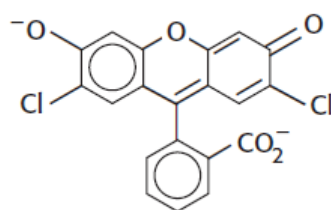
(b) 终点及之后, Ag^+ 过剩, 沉淀表面正电荷多, 吸附负电荷指示剂显色

FIGURE 26-13 Ions from solution are adsorbed on the surface of a growing crystallite. (a) A crystal growing in the presence of excess lattice anions (anions that belong in the crystal) will have a slight negative charge because anions are predominantly adsorbed. (b) A crystal growing in the presence of excess lattice cations will have a slight positive charge and can therefore adsorb a negative indicator ion. Anions and cations in the solution that do not belong in the crystal lattice are less likely to be adsorbed than are ions belonging to the lattice. These diagrams omit other ions in solution. Overall, each solution plus its growing crystallites must have zero total charge.

Fajans法常用吸附指示剂

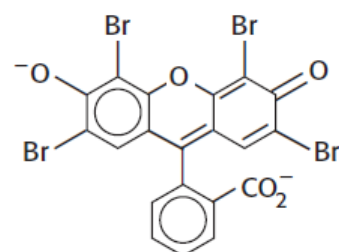
指示剂	pK_a	测定对象	滴定剂	颜色变化	滴定条件 (pH)
荧光黄	~7	Cl^- , Br^- , I^-	Ag^+	黄绿 — 粉红	7~10
二氯 荧光黄	~4	Cl^- , Br^- , I^-	Ag^+	黄绿 — 粉红	4~10
曙红	~2	Br^- , I^- , SCN^-	Ag^+	粉红 — 红紫	2~10
甲基紫		Ag^+	Cl^-	红 — 紫	酸性

Fajans法常用指示剂结构



Dichlorofluorescein

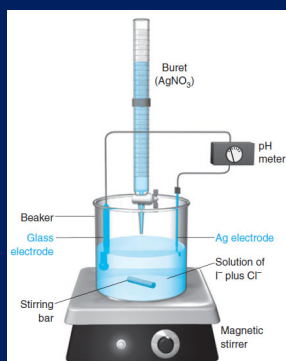
二氯荧光黄



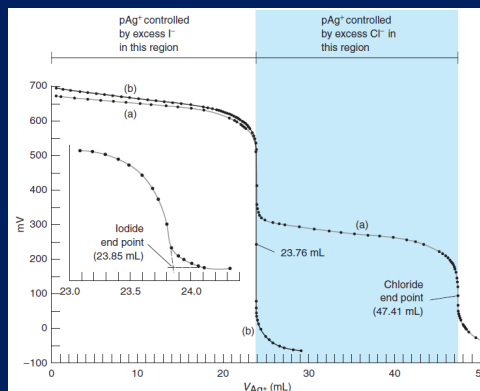
Tetrabromofluorescein (eosin)

四溴荧光黄（曙红）

电位法测定滴定曲线



Apparatus for measuring the titration curves in Figure 26-10. The silver electrode responds to changes in Ag^+ concentration, and the glass electrode provides a constant reference potential in this experiment. The measured voltage changes by approximately 59 mV for each factor-of-10 change in $[\text{Ag}^+]$. All solutions, including AgNO_3 , were maintained at pH 2.0 by using 0.010 M sulfate buffer prepared from H_2SO_4 and KOH.



Experimental titration curves.
(a) 40.00 mL of 0.0502 M KI plus 0.0500 M KCl titrated with 0.0845 M AgNO_3 . Inset expands the region near the first equivalence point. (b) 20.00 mL of 0.1004 M I^- titrated with 0.0845 M Ag^+ .

仪器分析法例子

电感耦合等离子体质谱仪 (ICP-MS)



<https://www.thermofisher.cn/cn/zh/home/virtual/icap-rqplus-icap-ms-3d-tour.html>

沉淀滴定的误差计算

例：以 $5.0 \times 10^{-3} \text{ M K}_2\text{CrO}_4$ 为指示剂，用 0.10 M AgNO_3 滴定 $50 \text{ ml } 0.10 \text{ M NaCl}$ ，计算滴定误差。

化学计量点 $[\text{CrO}_4^{2-}] = 2.5 \times 10^{-3} \text{ M}$

终点 $[\text{Ag}^+] = \sqrt{\frac{K_{sp}(\text{Ag}_2\text{CrO}_4)}{[\text{CrO}_4^{2-}]}} = \sqrt{\frac{1.9 \times 10^{-12}}{2.5 \times 10^{-3}}} = 2.76 \times 10^{-5} \text{ M}$

$[\text{Ag}^+]_{\text{溶解}} = [\text{Cl}^-]_{\text{溶解}} = K_{sp} / 2.76 \times 10^{-5} = 6.5 \times 10^{-6} \text{ M}$

过量 $[\text{Ag}^+]$ 等于过量加入部分减去 AgCl 溶解部分

$[\text{Ag}^+]_{\text{过量}} = 2.76 \times 10^{-5} - 6.5 \times 10^{-6} = 2.11 \times 10^{-5} \text{ M}$

滴定误差

$$TE\% = \frac{2.11 \times 10^{-5} \times 100}{0.10 \times 50} \times 100 = 0.04\%$$

本章小结

1. 溶度积 K_{sp} 、活度积 K_{ap} 的定义，与溶解度的关系
2. 同离子效应、酸效应、配位效应、离子强度对溶解度的影响；条件溶度积
3. 重量分析计算
4. 摩尔法、佛尔哈德法、法扬司法的基本原理
5. 沉淀滴定各阶段及滴定误差计算

本章作业

p155-157

思考题 3, 5, 6(2), 8

习题 2, 4, 5, 8, 10